

Project Design Phase
Solution Architecture

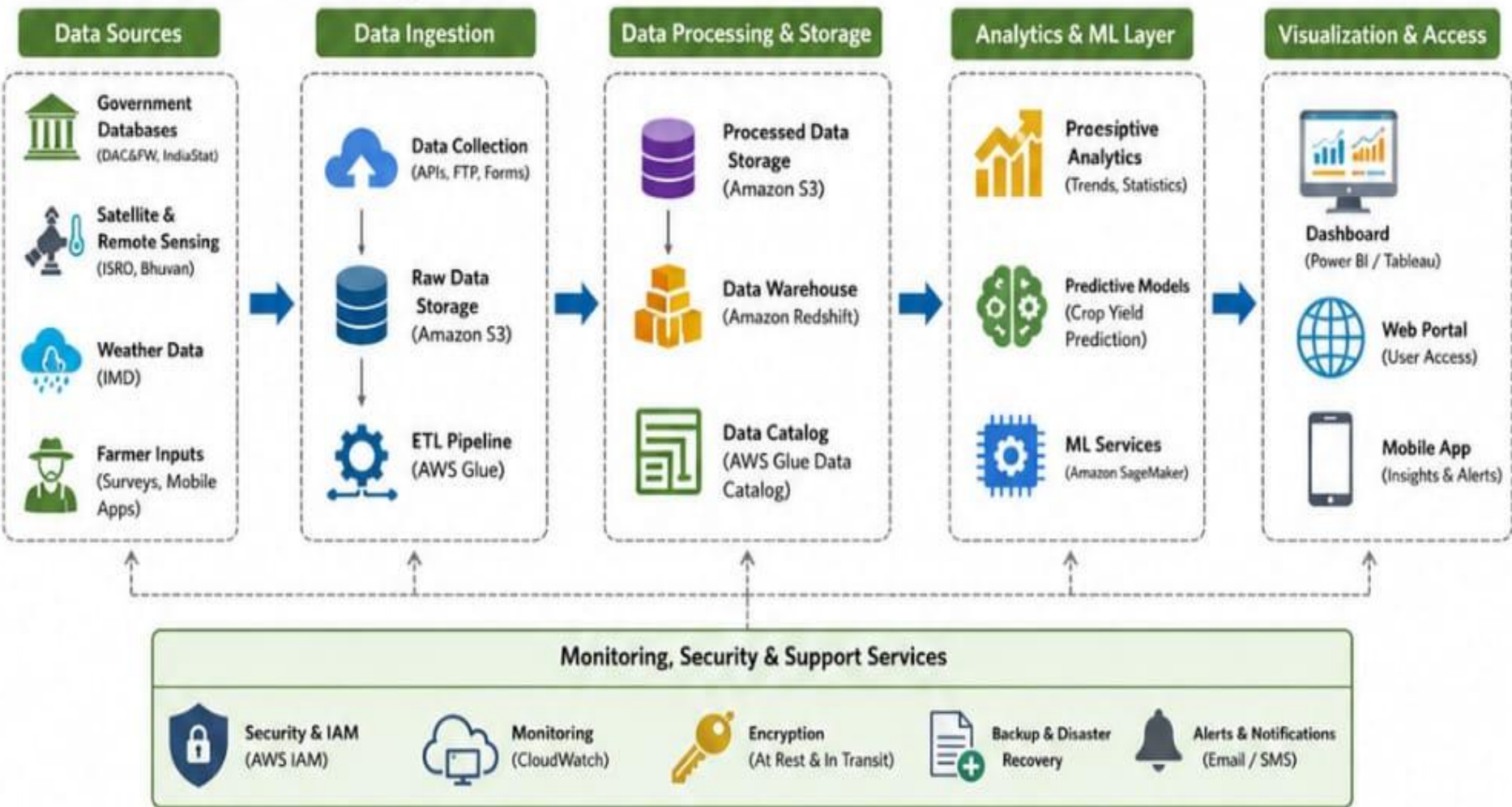
Date	15 February 2025
Team ID	AI-ML-25-07
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture for India’s Agricultural Crop Production Analysis project defines how agricultural data is collected, processed, analyzed, and visualized to provide insights for better decision-making. It bridges the gap between business needs and technology by integrating data sources, analytics pipelines, and user-facing dashboards.

- Find the best tech solution:** Utilize big data, cloud computing, and machine learning to analyze crop production trends, regional performance, and yield predictions.
- Describe the structure, characteristics, behavior:** Data flows from multiple agricultural sources through ETL pipelines to analytics and visualization layers, ensuring scalability, reliability, and security.
- Define features, development phases, and requirements:** Include data collection, data processing, analysis, dashboard development, and deployment with clear functional and non-functional requirements.
- Provide specifications:** Define data standards, APIs, security policies, and performance metrics to ensure the solution is effectively designed, managed, and delivered.

Solution Architecture Diagram:



- Key Benefits:**
- ✓ Comprehensive crop production insights across India
 - ✓ Data-driven decision making for farmers, policymakers & researchers
 - ✓ Scalable, secure, and reliable cloud-based architecture